Introduction to Machine Learning

Chapter 1: What is Machine Learning?
Machine Learning (ML) is a subset of Artificial Intelligence that focuses on building applications that lea

Chapter 2: Three Main Paradigms of Machine Learning
1. Supervised Learning: The model trains on labeled data (e.g. classification of spam emails, house pri
2. Unsupervised Learning: The model finds hidden patterns in unlabeled data (e.g. customer clustering
3. Reinforcement Learning: An agent learns by interacting with an environment and receiving rewards o

Chapter 3: The Machine Learning Pipeline
A typical ML workflow consists of:
- Data Collection and Preparation
- Feature Engineering
- Model Selection and Training
- Evaluation and Fine-Tuning
- Deployment and Monitoring